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Chapter 13

Cisco Device Security Best Practices

The CCNA certification expects IT professionals to have a working knowledge of how to configure a Cisco router and switch in a secure manner. CCNA certified professionals are expected to be able to configure proper banners, passwords, port security, and communication protocols. This chapter tests you on these issues and more.

The Problems You'll Work On

In this chapter, you'll review questions concerning the following topics:

- Using strong passwords and encrypting passwords in the configuration
- Configuring port security on your switches to limit which systems can connect to the port
- Using communication protocols that encrypt communication
- Creating usernames and passwords that are used to log in to the device

What to Watch Out For

Don't let common mistakes trip you up; watch for the following when working with these questions:

When answering questions for the CCNA exam(s), watch for choices involving security features that should be used. Here are a few key points to remember when answering security-related questions:

- Use strong passwords and encrypt passwords.
- Ensure that after the password is configured for a port, the login command is used to force the device to prompt for a password.
- If you have configured usernames and passwords for the device, ensure that you have used the login local command so that the device prompts for both a username and password.
- Always encrypt communication with SSH when possible.

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Security Basics

466–473 Choose the best answer(s).

466. What type of attack involves the hacker overloading a system and causing the system to crash or not perform its job role?

(A) Buffer overflow attack

(B) DoS attack

(C) Dictionary attack

(D) MiTM attack

467. You would like to control communication between the accounting and marketing groups. What feature of a Cisco switch will allow you to do this?

(A) CDP

(B) Port mirroring

(C) VLANs

(D) Port security

468. What network technology could be used to encrypt communication from point A to point B over an untrusted network?

(A) VPN

(B) VLAN

(C) NAP

(D) NAT

469. What type of firewall can filter traffic by analyzing the payload of a packet?

(A) Stateful packet inspection firewall

(B) Packet filtering firewall

(C) Application layer firewall

(D) Network address translation

470. What type of firewall filters traffic based on the context of the conversation and the layer-3 and layer-4 header?

(A) Packet filtering firewall

(B) Application layer firewall

(C) Proxy server

(D) Stateful packet inspection firewall

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471. What type of attack involves the hacker altering the source IP address of a packet in order to bypass your packet filtering rules on the router?

(A) MiTM

(B) Spoofing

(C) DoS

(D) Buffer overflow

472. Your manager has asked you make sure that the Cisco router configuration files are protected from the intruders on the Internet. Which actions would you recommend? (Select two.)

(A) Only allow unrestricted access to the VTY ports.

(B) Use a firewall to control access to the device from the Internet.

(C) Use Telnet to remotely configure the device.

(D) Use SSH to remotely configure the device.

473. You are helping the security team create a network security plan for the business. Which of the following should you include in the plan?

(A) Disable antivirus automatic updates in order to reduce traffic.

(B) Lock network equipment in a secure location.

(C) Ensure that all Telnet ports use the password of telnet.

(D) Ensure to use the no service-password encryption command.

Security Configuration

474–479 Choose the best answer(s).

474. You have been asked to review the running-configuration of a customer's router and identify potential security issues related to the configuration. Review the output of the `show running-config` command, shown below. Which of the following are potential security issues with this configuration? (Select two.)

```
ROUTERB#show running-config
Building configuration…

(output omitted for briefness)
service timestamps log datetime msec
no service password-encryption
!
hostname ROUTERB
!
ip subnet-zero
```

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```
!
interface FastEthernet0/1
  ip address 27.0.0.1 255.0.0.0
  no ip directed-broadcast
!
interface Serial0/0/0
  ip address 26.0.0.2 255.0.0.0
  no ip directed-broadcast
!
router rip
  network 26.0.0.0
  network 27.0.0.0
!
banner motd
Welcome to Glen's router! You have connected to 192.168.0.1!
!
line con 0
  password myc0nP@ss
  login

line aux 0
line vty 0 4
  password telnet
  login
end
```

(A) The console password is not a complex password.

(B) Passwords are not encrypted.

(C) ip subnet zero is being used.

(D) The banner is using inappropriate text.

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475. The junior IT technician has configured the router for usernames and passwords, but the router is not prompting for usernames and passwords when connecting to the console port. You review the configuration, shown below. What is the problem?

```
ROUTERB#show running-config

Building configuration…

(output omitted for briefness)
no service password-encryption
!
hostname ROUTERB
!
ip subnet-zero
!
Username bob password P@ssw0rd
Username sue password Pa$$w0rd
!
interface FastEthernet0/1
  ip address 27.0.0.1 255.0.0.0
  no ip directed-broadcast
!
interface Serial0/0/0
  ip address 26.0.0.2 255.0.0.0
  no ip directed-broadcast
!
line con 0
  login

line aux 0
line vty 0 4
  password telnet
  login
end
```

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(A) The console password is missing.

(B) Passwords are not encrypted.

(C) ip subnet zero is being used.

(D) The login local command should be used.

476. Sue has configured the console port with a password, but it does not seem to work. You review the configuration, shown below. What is causing the problem?

```
ROUTERB#show running-config

Building configuration…

(output omitted for briefness)
!
hostname ROUTERB
!
ip subnet-zero
!
interface FastEthernet0/1
  ip address 27.0.0.1 255.0.0.0
  no ip directed-broadcast
!
interface Serial0/0/0
  ip address 26.0.0.2 255.0.0.0
  no ip directed-broadcast
!
line con 0
  password Con$0le1

line aux 0
line vty 0 4
end
```

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(A) The console password is not complex.

(B) Passwords are not encrypted.

(C) The `login` command is missing.

(D) The `login local` command is missing.

477. For security reasons, you have decided to disable the built-in web server on the Cisco device. Which command would you use?

(A) R1(config)# `no ip http server`

(B) R1(config)# `disable ip http server`

(C) R1(config)# `ip http server off`

(D) R1(config)# `ip http server disabled`

478. You would like to create a user named Bob that has privilege level 3. What command would you use?

(A) R1(config)# `username Bob password P@ssw0rd level 3`

(B) R1(config)# `username Bob privilege 3 password P@ssw0rd`

(C) R1# `username Bob privilege 3 password P@ssw0rd`

(D) R1# `username Bob password P@ssw0rd level 3`

479. How would you view your current privilege level on a Cisco device?

(A) `display privilege`

(B) `display level`

(C) `show level`

(D) `show privilege`

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Switch Security

480–488 Choose the best answer(s).

480. What command is used to disable a port on the switch?

(A) switch(config)#`shutdown port 5`

(B) switch(config-if)#`shutdown`

(C) switch(config)#`port 5 shutdown`

(D) switch(config-if)#`disable`

481. You have been asked to configure port number 4 to only allow the system that is currently connected to the port access to the network via that port. What Cisco switch feature would you use?

(A) Maximum MAC addresses

(B) MAC address table

(C) STP

(D) Sticky

482. Which violation mode would you use to disable a port when an unauthorized device connects to the port, and have the port disabled until the administrator enables it?

(A) Restrict

(B) Shutdown

(C) Protect

(D) Disable

483. Which feature allows you to control which workstations can connect to a specific port on the switch?

(A) Port mirroring

(B) VLANs

(C) Port security

(D) MAC address table

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484. You are trying to connect a system to a port on the switch, but having no luck. You use the `show interface f0/5` command to view the status of the port and receive the output shown below. What should you do to allow the system to connect via the port?

```
SW1#show interface f0/5
FastEthernet0/5 is administratively down, line protocol is down (disabled)
  Hardware is Lance, address is 0060.2fba.e405 (bia 0060.2fba.e405)
BW 100000 Kbit, DLY 1000 usec,
  reliability 255/255, txload 1/255, rxload 1/255
Encapsulation ARPA, loopback not set
Keepalive set (10 sec)
Half-duplex, 100Mb/s
```

(A) Change the MAC on the port.

(B) Change the IP on the port.

(C) Alter the MAC address table.

(D) Enable the port.

485. You wish to limit which device can connect to each port on the switch by configuring port security with the MAC address of each system currently connected to the switch. What command would you use?

(A) SW1(config-if)#`switchport port-security mac-address sticky`

(B) SW1(config-if)#`switchport port-security`

(C) SW1(config-if)#`mac-address sticky`

(D) SW1(config-if)#`port-security mac-address automatic`

486. Which of the following port modes supports the port security feature?

(A) secure

(B) trunk

(C) access

(D) workstation

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487. You would like to configure port 5 on your switch to allow only one system to connect to the port that uses the MAC address of 1111.2222.3333. Which of the following configurations would you use?

(A) Use the following command:

```
SW1(config)#switchport mode access
SW1(config)#switchport port-security
SW1(config)#switchport port-security mac-address 1111.2222.3333
SW1(config)#switchport port-security maximum 1
SW1(config)#switchport port-security violation shutdown
```

(B) Use the following command:

```
SW1(config)#interface f0/5
SW1(config-if)#switchport mode access
SW1(config-if)#switchport port-security
SW1(config-if)#switchport port-security mac-address 1111.2222.3333
SW1(config-if)#switchport port-security maximum 1
SW1(config-if)#switchport port-security violation shutdown
```

(C) Use the following command:

```
SW1(config)#interface f0/5
SW1(config-if)#switchport mode access
SW1(config-if)#switchport port-security
SW1(config-if)#switchport port-security mac-address sticky
SW1(config-if)#switchport port-security maximum 1
SW1(config-if)#switchport port-security violation shutdown
```

(D) Use the following command:

```
SW1(config)#interface f0/5
SW1(config-if)#switchport mode trunk
SW1(config-if)#switchport port-security
SW1(config-if)#switchport port-security mac-address 1111.2222.3333
SW1(config-if)#switchport port-security maximum 1
SW1(config-if)#switchport port-security violation shutdown
```

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488. Your manager would like to use a single system on the network for monitoring network traffic. You have connected the monitoring system to port 12 on the switch, but you are unable to monitor network traffic. What do you need to do? (Select two.)

(A) Connect the monitoring system to port 1 on the switch.

(B) Configure port security.

(C) Configure VLANs on the switch.

(D) Configure port mirroring on the switch.

Configuring Passwords

489–494 Choose the best answer(s).

489. You wish to have the Cisco device prompt you for a username and password. What command would you use?

(A) router(config)#`login local`

(B) router(config-line)#`login local`

(C) router(config)#`login`

(D) router(config-line)#`login`

490. Your manager would like all passwords encrypted in the configuration files on the Cisco routers and switches. What command is used to encrypt all passwords in the configuration files?

(A) router#`service password-encryption`

(B) router(config)#`enable service password-encryption`

(C) router#`enable service password-encryption`

(D) router(config)#`service password-encryption`

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491. Which of the following commands are used to configure a password on your console port and ensure that the router prompts for a password anytime someone connects to that port?

(A) Use the following commands:

ROUTERA(config)#`password con`

ROUTERA(config)#`login`.

(B) Use the following commands:

ROUTERA(config)#`line vty 0 4`

ROUTERA(config-line)#`password con`

ROUTERA(config-line)#`login`.

(C) Use the following commands:

ROUTERA(config)#`line con 0`

ROUTERA(config-line)#`password con`

ROUTERA(config-line)#`login`.

(D) Use the following commands:

ROUTERA(config)#`line con 0`

ROUTERA(config-line)#`password con`.

492. You wish to create a username `Rebecca` with a password of `mypass`, what command would you use?

(A) router(config)#`enable user rebecca password mypass`

(B) router(config-if)#`username rebecca password mypass`

(C) router(config)#`create user rebecca password mypass`

(D) router(config)#`username rebecca password mypass`

493. You wish to configure a password for priv exec mode but have the password encrypted in the configuration file. What command would you use?

(A) `enable secret P@ssw0rd`

(B) `enable password P@ssw0rd`

(C) `enable encrypted password P@ssw0rd`

(D) `enable secure password P@ssw0rd`

494. What is the difference between the login command and the login local command?

(A) Login prompts for a username and password, whereas login local prompts for a password.

(B) Login is used for Telnet access, whereas login local is used for access using the local console port.

(C) Login is used for local console port access, whereas login local is used for Telnet access.

(D) Login prompts for a password, whereas login local prompts for a username and password.

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Configuring Banners

495–497 Choose the best answer(s).

495. You are configuring banners on your Cisco router. Which of the following banners is displayed last?

(A) The exec banner

(B) The login banner

(C) The MOTD banner

(D) All banners are shown at the same time.

496. Which of the following commands configures a proper message of the day banner?

(A) Use the following commands:

R1(config)#`banner motd #`

R1(config)#`This device is for authorized individuals only.`

`$`

(B) Use the following commands:

R1#`banner motd #`

R1#`This device is for authorized individuals only.`

`#`

(C) Use the following commands:

R1(config)#`banner motd #`

R1(config)#`This device is for authorized individuals only.`

`#`

(D) Use the following commands:

R1(config)# `motd banner`

R1(config)#`This device is for authorized individuals only.`

497. Which of the following banner type displays before the user is asked to log in, but after the MOTD banner?

(A) Exec

(B) MOTD

(C) Logout

(D) Login

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Remote Management

498–503 Choose the best answer(s).

498. Your manager would like to ensure that remote access to the router is only allowed using a secure communication protocol. Which command would you use?

(A) router(config)#`transport input ssh`

(B) router(config-line)#`transport input ssh`

(C) router(config-line)#`transport input both`

(D) router(config-line)#`transport input telnet`

499. You are responsible for configuring Telnet access to the router. Which of the following commands would you use to ensure a username and password are required for Telnet access?

(A) Use the following commands:

ROUTERA(config)#`username bob password pass`

ROUTERA(config)#`line con 0`

ROUTERA(config-line)#`login local`.

(B) Use the following commands:

ROUTERA(config)#`username bob password pass`

ROUTERA(config)#`line vty 0 4`

ROUTERA(config-line)#`login`.

(C) Use the following commands:

ROUTERA#`username bob password pass`

ROUTERA#`config term`

ROUTERA(config)#`line vty 0 4`

ROUTERA(config-line)#`login`.

(D) Use the following commands:

ROUTERA(config)#`username bob password pass`

ROUTERA(config)#`line vty 0 4`

ROUTERA(config-line)#`login local`.

500. You want to remotely manage your Cisco devices. What protocol should you use?

(A) Telnet

(B) SSH

(C) HTTP

(D) FTP

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501. You are having trouble telnetting into a router from your client system. You can successfully ping the IP address of the router. What may be the problem?

(A) The default gateway is incorrect.

(B) There is no Telnet password.

(C) The subnet mask on the client is incorrect.

(D) The IP address on the client is on the wrong subnet.

502. Which of the following commands are used to configure the router to only support SSH for remote management?

(A) Use the following commands:

```
R1(config)#crypto key generate rsa
R1(config)#line vty 0 4
R1(config-line)#login local
R1(config-line)#transport input ssh
```

(B) Use the following commands:

```
R1(config)#crypto key generate rsa
R1(config)#line vty 0 4
R1(config-line)#login local
```

(C) Use the following commands:

```
R1(config)#crypto key generate rsa
R1(config)#line vty 0 4
R1(config-line)#transport input telnet
```

(D) Use the following commands:

```
R1(config)#crypto key generate rsa
R1(config)#line ssh 0 4
R1(config-line)#login local
R1(config-line)#transport input ssh
```

503. What command would you use to configure the Telnet ports to support both SSH and Telnet as protocols?

(A) `transport input ssh`

(B) `transport input telnet`

(C) `transport input ssh telnet`

(D) `transport input http`